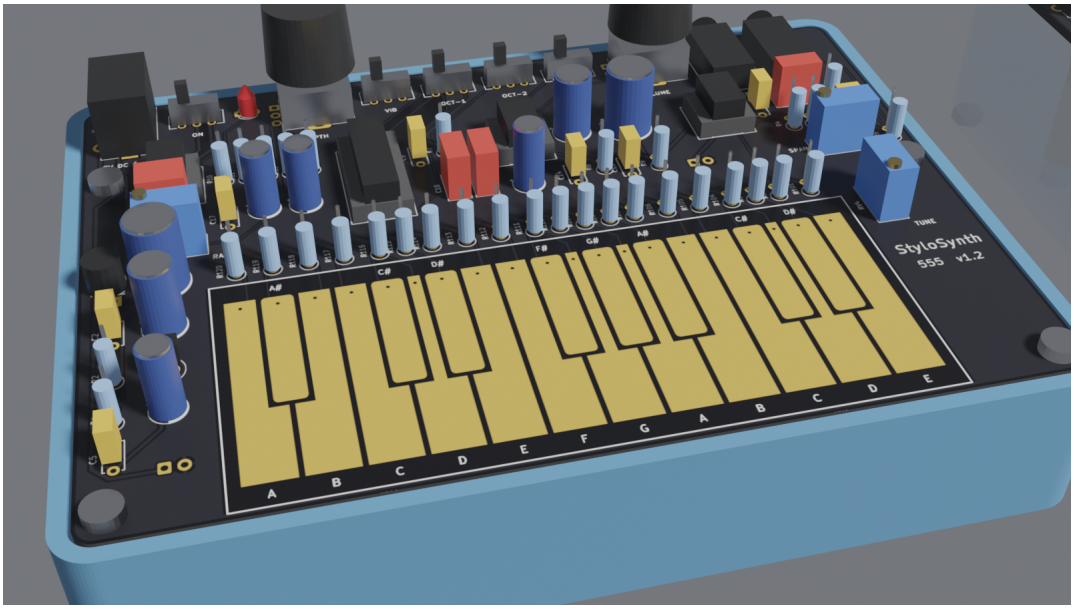
StyloSynth-555

A 555-based stylus synthesizer

Design & Verification Report



05 October 2026

Design revision v1.2 · commit `ff20aea`

Abstract. StyloSynth-555 is a stylophone-style analogue synthesizer with a CMOS 555 oscillator, a 20-key exposed-copper keyboard, four exact octaves from a binary divider, an LFO vibrato and an LM386 amplifier. It runs on a single 136×98 mm two-layer PCB powered from a 9 V battery. This report describes the design and presents the automated verification: circuit simulation, a 2000-build tolerance analysis, schematic/netlist consistency, static electrical-rule checking, power budgets, PCB DRC and mechanical fit. All 6 verification steps pass. The design has not yet been validated on hardware.

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1 Introduction

The instrument is inspired by the vintage Stylophone and by the TouchTone555 project. None of the reference files are reused; this is an independent design that keeps the feature set and fixes the issues the reference documented:

- **Pitch drift with volume** → the oscillator gets its own LDO-regulated, RC-filtered 5 V rail.
- **Inexact octaves from capacitor banks** → a CD4040 binary divider gives mathematically exact 2:1 ratios.
- **Op-amp unable to drive the speaker** → an LM386 power amplifier.
- **20 tuning trimmers** → fixed 1 % resistors plus a two-trimmer (TUNE/SPAN) calibration, verified statistically in Section 5.

The design is fully script-generated. `circuit.py` is the single source of truth for the schematic, PCB and BOM, and `verify_all.py` re-runs every check in this report.

2 System architecture

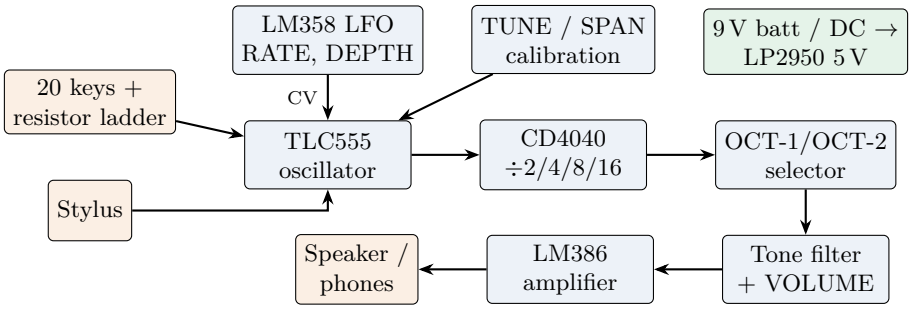


Figure 1: Signal flow. The oscillator, divider and LFO run on a regulated 5 V rail; the amplifier runs from the raw 9 V supply.

- **Oscillator.** A TLC555 in the 50 %-duty topology: the output charges the 22 nF polypropylene timing capacitor through the key ladder. The stylus picks a tap, so the timing resistance and the pitch depend on the key touched. A 10 MΩ pull-down silences the oscillator when the stylus is lifted.
- **Octaves.** The CD4040 divides by 2, 4, 8 or 16. Two DPDT switches form a 2-bit selector.
- **Vibrato.** An LM358 relaxation oscillator (3.9–9.6 Hz) is buffered and AC-coupled through the DEPTH pot into the 555 control-voltage pin.
- **Audio.** A series resistor and switchable tone capacitor feed the VOLUME pot and an LM386 at gain 20, with a Zobel network and a 220 µF output capacitor. The headphone jack's switch contacts disconnect the speaker and add a 220 Ω series resistor.
- **Power.** 9 V battery or 9 V centre-negative DC (the jack switch disconnects the battery), Schottky reverse protection, and an LP2950-5.0 LDO behind a 47 Ω/100 µF filter.

3 Circuit design

3.1 Oscillator and key ladder

For a rail-to-rail 555 whose thresholds are V_{CV} and $V_{CV}/2$, the capacitor charges towards V_{DD} and discharges towards 0, so

$$f \;=\; \frac{1}{R\,C_T\left[\ln\dfrac{V_{DD}-V_{CV}/2}{V_{DD}-V_{CV}}+\ln 2\right]}, \qquad R = R_0 + \sum_{i\le k} R_i, \tag{1}$$

where R_0 is the 555 output resistance plus the 100 Ω stylus resistor. With $C_T = 22$ nF and $V_{CV} = 3.296$ V, the 555 spans 880–2637.02 Hz. Segment values are chosen greedily from E96 so that the *cumulative* resistance tracks each target, which means rounding errors never accumulate.

Calibration. TUNE (RV4, 2000 Ω) adds an offset common to all keys and is set on key 20. SPAN (RV5, 1000 Ω) sits in the CV divider: it changes V_{CV}, which scales every frequency by the same factor, and is set on key 1. Two adjustments cancel the two dominant unknowns (R_0 and the C_T tolerance).

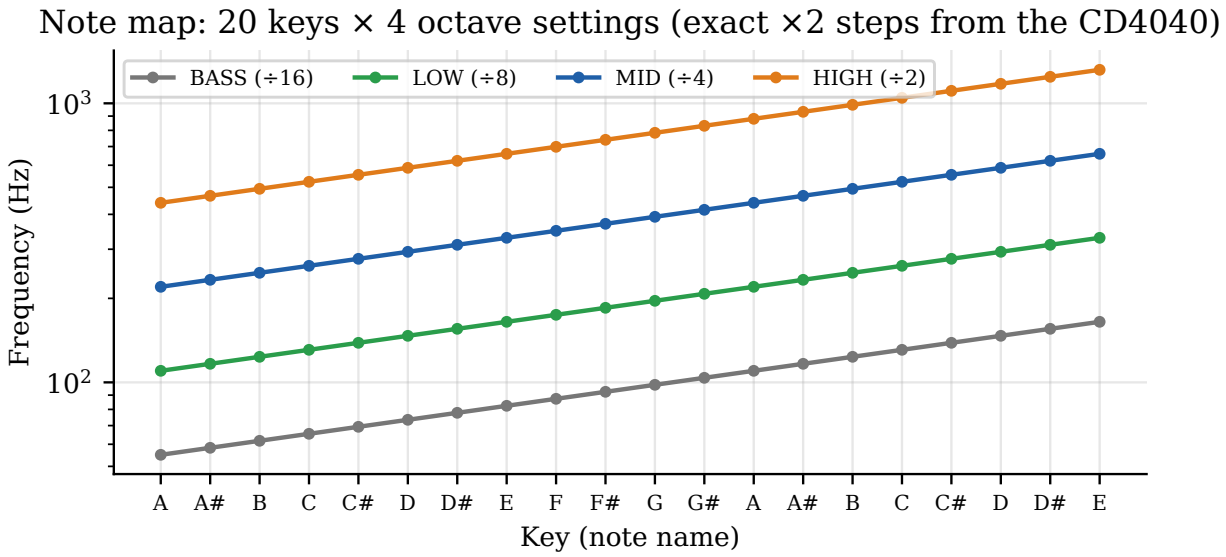


Figure 2: Design frequency of every key at each octave setting.

Table 1: Ladder values (key 20 is fed first from the 555 output).

Key	Note (MID)	Segment R (Ω)	f_{555} (Hz)	Output, MID (Hz)
1	A3	2k15	880.0	220.0
2	A#3	2k	932.3	233.1
3	B3	1k87	987.8	246.9
4	C4	1k78	1046.5	261.6
5	C#4	1k69	1108.7	277.2
6	D4	1k58	1174.7	293.7
7	D#4	1k5	1244.5	311.1
8	E4	1k4	1318.5	329.6
9	F4	1k33	1396.9	349.2
10	F#4	1k27	1480.0	370.0
11	G4	1k18	1568.0	392.0
12	G#4	1k13	1661.2	415.3
13	A4	1k05	1760.0	440.0
14	A#4	1k	1864.7	466.2
15	B4	953	1975.5	493.9
16	C5	887	2093.0	523.2
17	C#5	845	2217.5	554.4
18	D5	787	2349.3	587.3
19	D#5	750	2489.0	622.2
20	E5	11k3 + TUNE	2637.0	659.3

3.2 Vibrato, audio and power

The LFO is a single-supply Schmitt relaxation oscillator with $3 \times 100\,\text{k}\Omega$ hysteresis and $R = 68\,\text{k}\Omega$+ RATE ($100\,\text{k}\Omega$) into $1\,\mu\text{F}$; its timing node is buffered by the second op-amp half. A $1.5\,\text{k}\Omega$ / ($2.4\,\text{k}\Omega$ + SPAN) external CV divider swamps the TLC555's internal divider, so the vibrato depth and the calibration don't depend on the 555's internal resistors. In the audio path, a $150\,\text{k}\Omega$ series resistor sets the LM386 input so full volume just avoids hard clipping at 9 V.

4 Simulation results

All circuit simulations use ngspice (the KiCad 7 shared library) driven from Python. The 555 is modelled behaviourally as a CV-referenced Schmitt switch with $100\,\Omega$ output resistance, and the LM358 as a saturating high-gain stage (0.02–3.5 V swing at 5 V).

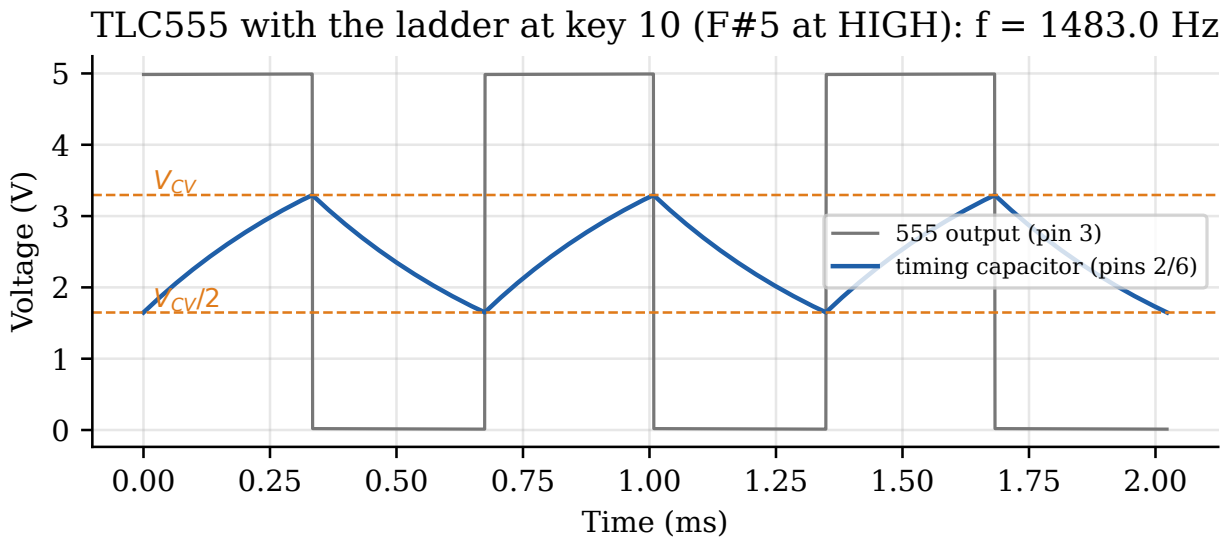


Figure 3: Oscillator transient: the timing capacitor swings between $V_{CV}/2$ and V_{CV}.

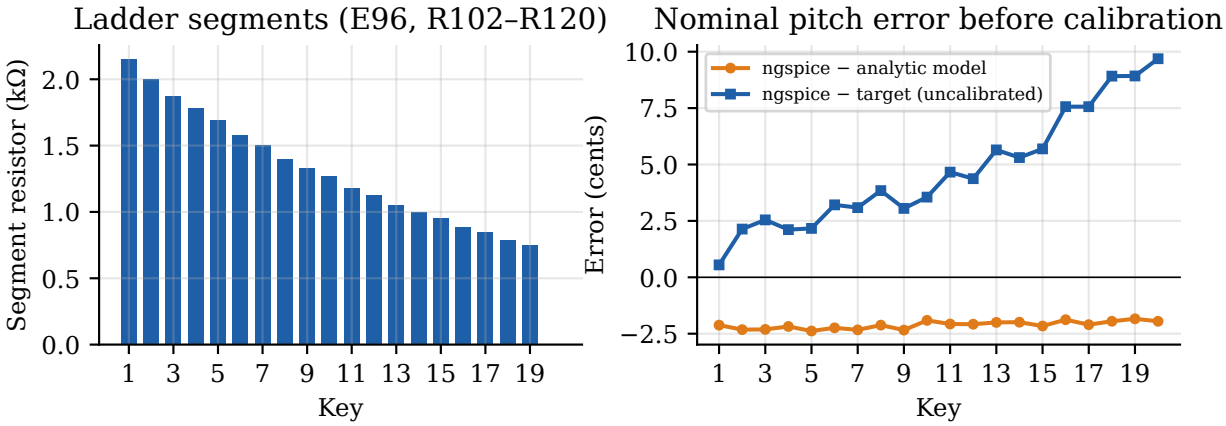


Figure 4: Ladder segment values and nominal (uncalibrated) pitch error. The near-constant offset between simulation and target is removed by TUNE/SPAN; the simulation agrees with the analytic model to within 0.54 cents across all keys.

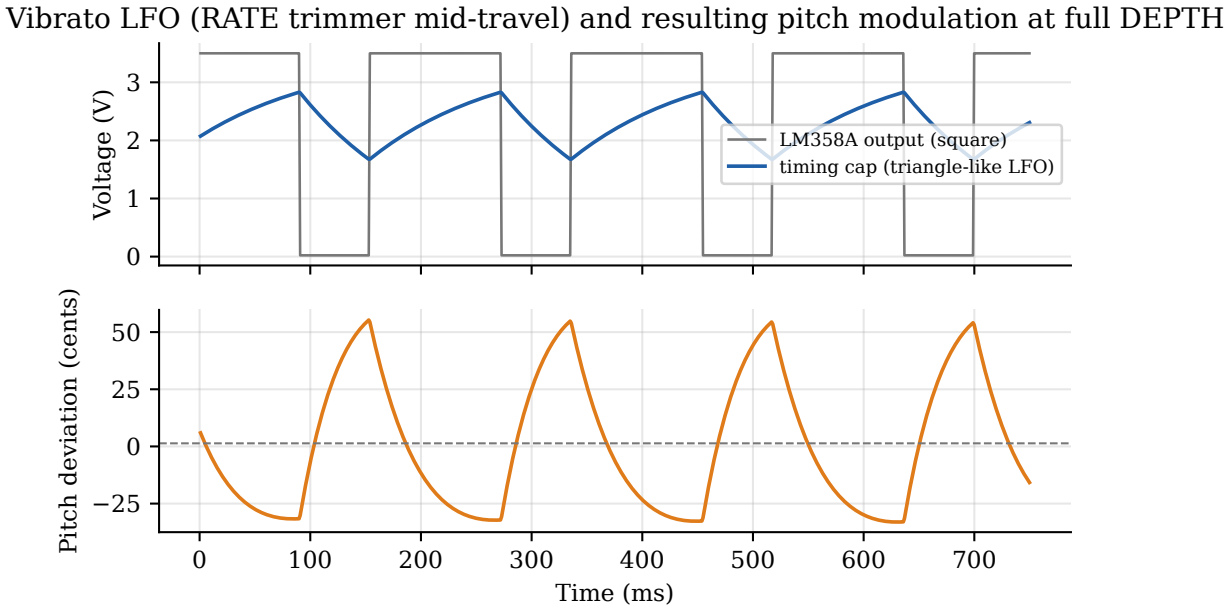
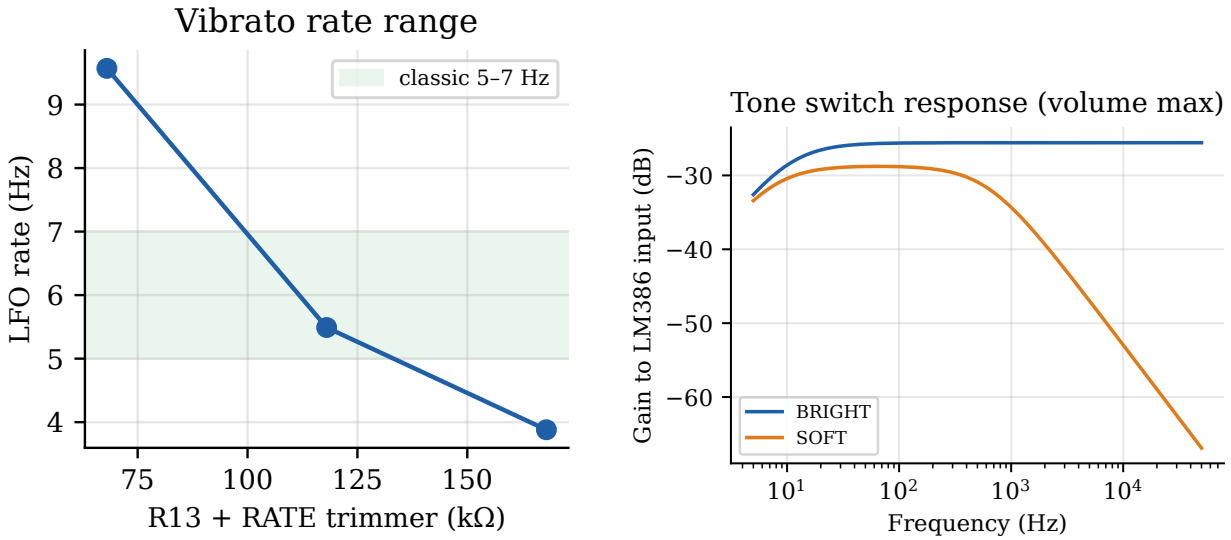


Figure 5: Vibrato LFO waveform and pitch modulation at full DEPTH: -33 to +55 cents. The LM358 output swing is asymmetric (0.02–3.5 V), so the time-averaged pitch rises by 1 cents while vibrato is on (dashed line). The offset shrinks in proportion to DEPTH; calibration is done with vibrato off.



(a) LFO rate vs. RATE setting

(b) Tone response into the LM386

Figure 6: Vibrato rate range and audio path.

Table 2: Simulation checks.

Check	Target	Result	Status
Ladder: ngspice vs analytic model	spread < 2 ct	0.54 ct	**PASS**
LFO rate range	covers 5–7 Hz	3.88–9.57 Hz	**PASS**
Vibrato depth (full)	25–80 ct	±45.6 ct	**PASS**
CV DC with LFO running	3.296 V	3.2956 V	**PASS**
LM386 drive, full volume	2.0–3.8 V pk	3.35 V pk	**PASS**
Audio HPF corner	< 40 Hz	10.4 Hz	**PASS**
SOFT LPF corner	0.5–1 kHz	629.5 Hz	**PASS**

5 Tolerance analysis

Each simulated build draws random values for: every ladder resistor (±1 % or ±0.1 %), C_T (±2.5 %), R_0 (100–700 Ω) and the 555 internal divider (20–300 kΩ). The two-point calibration is then emulated (TUNE on key 20, SPAN on key 1, iterated) and the error of every key is measured. The model behind this is the analytic equation, validated against ngspice in the previous section.

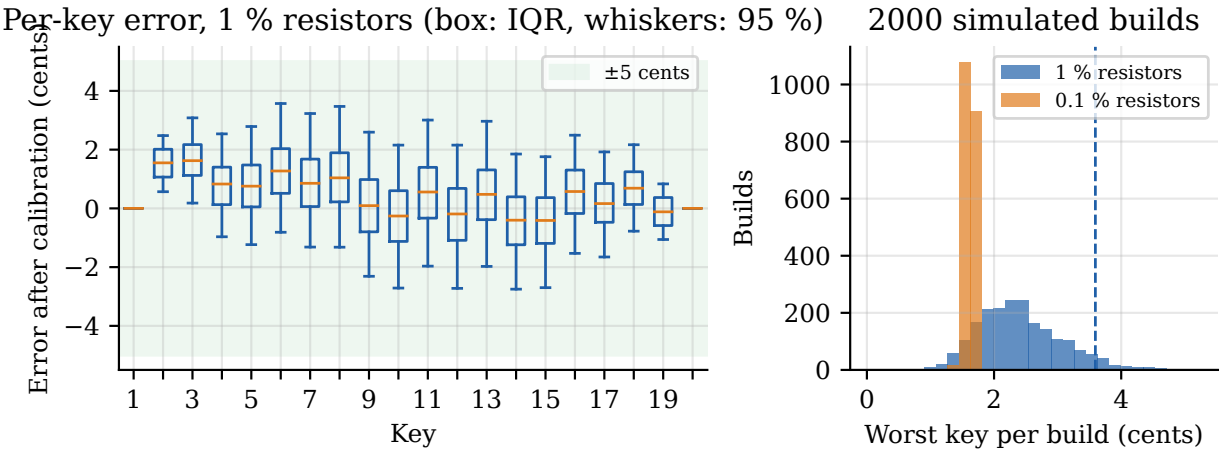


Figure 7: Tuning error after calibration across 2000 simulated builds.

Table 3: Worst-key tuning error per build (1 cent = 1/100 semitone; about 5–10 cents is the threshold of audibility).

Resistors	Cal. failures	Median worst key	95th pct	Max
1 % (specified)	0/2000	2.34 ct	**3.59 ct**	5.01 ct
0.1 % (upgrade)	0/2000	1.62 ct	1.75 ct	1.84 ct

6 Schematic and electrical rules

The schematic is generated from `circuit.py` and checked by round-trip: KiCad exports its netlist, which is compared with the design source. **Result:** PASS: 59 nets and 81 components match circuit.py.

KiCad 7 has no command-line ERC, so a static rule check runs on the netlist using the official KiCad symbol pin types. It covers 59 nets and 210 pins, and checks for: multiple drivers on one net, power inputs without a source, floating inputs and single-pin nets. **Result: 0 issues PASS**. Waived, intentional cases:

- U2 pin 7 unconnected - TLC555 DIS is an open-drain output (KiCad types it as input); unused in this topology
- U4 pin 1 unconnected - LM386 GAIN pins left open = gain 20 (datasheet default)
- U4 pin 8 unconnected - LM386 GAIN pins left open = gain 20 (datasheet default)

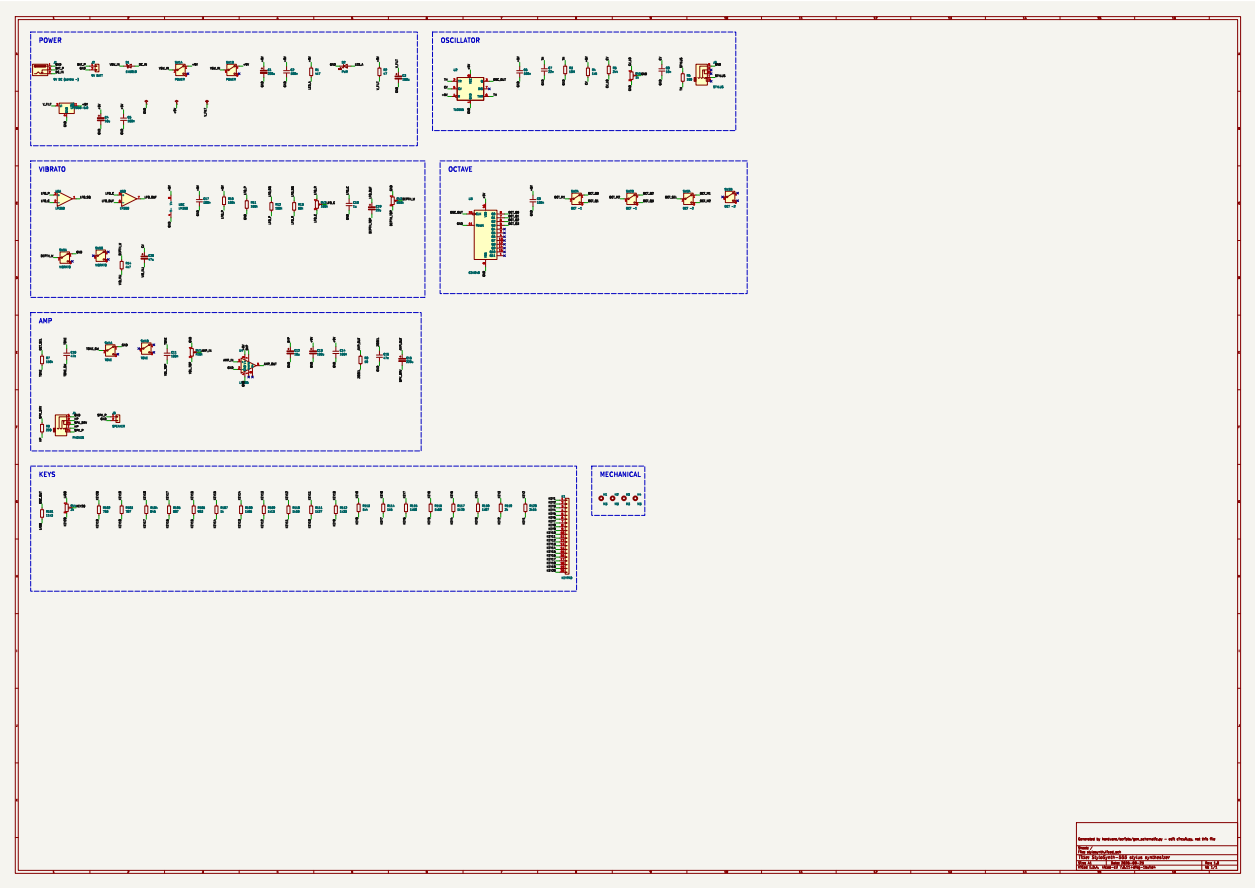


Figure 8: Generated schematic (vector; the full-size sheet is in Appendix B).

7 Power, ratings and dissipation

Datasheet-typical values, not measurements.

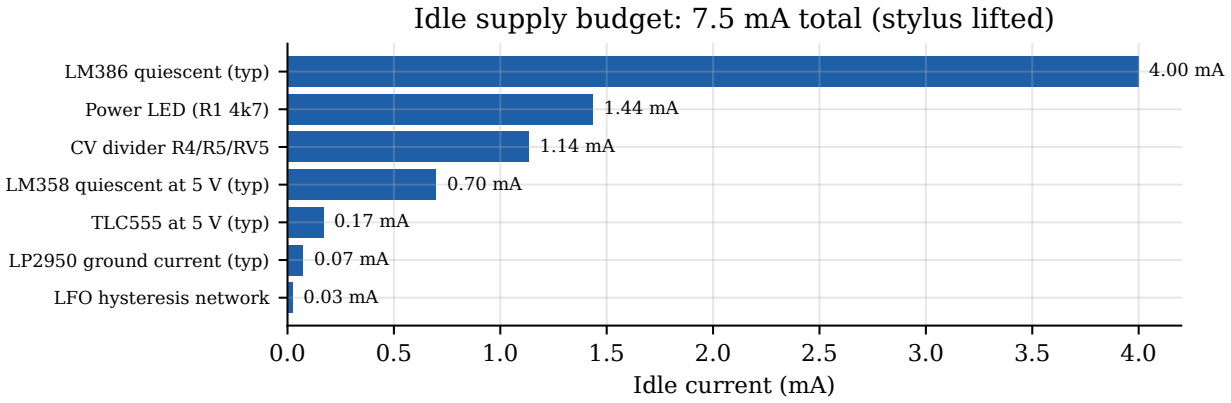


Figure 9: Idle current budget.

Table 4: Power and dissipation checks.

Item	Value	Limit / note	Status
Idle supply current	7.5 mA	≈ 73 h idle on a 550 mAh alkaline 9 V	**PASS**
LM386 output at full volume (sim)	3.35 V pk → 701 mW sine-equivalent into 8 Ω	use a ≥ 1 W speaker	**PASS**
LM386N-1 worst-case dissipation	549 mW	DIP-8 rating 1.25 W at 25 °C	**PASS**
LP2950 dissipation	10.7 mW at 3 mA	TO-92 rating ≈ 600 mW	**PASS**
R9 (220 Ω) with shorted headphone plug	51 mW	0.25 W rating	**PASS**
R1 LED resistor	9.7 mW	0.25 W rating	**PASS**
D1 1N5819 current / reverse voltage	< 0.3 A / ≤ 12 V	1 A / 40 V rating	**PASS**

Table 5: Capacitor voltage ratings (≥1.25× the worst-case net voltage; 12 V assumed on the 9 V side).

Cap	Value	Rated	Worst		Cap	Value	Rated	Worst	
C1	220u	16 V	12 V	**PASS**	C11	100n	63 V	5 V	**PASS**
C2	100n	50 V	12 V	**PASS**	C12	10u	25 V	12 V	**PASS**
C3	100u	16 V	12 V	**PASS**	C13	100u	16 V	12 V	**PASS**
C4	10u	25 V	5 V	**PASS**	C14	100n	50 V	12 V	**PASS**
C5	100n	50 V	5 V	**PASS**	C15	47n	50 V	5 V	**PASS**
C6	100n	50 V	5 V	**PASS**	C16	220u	16 V	12 V	**PASS**
C7	22n	100 V	5 V	**PASS**	C17	100n	50 V	5 V	**PASS**
C8	10n	50 V	5 V	**PASS**	C18	1u	63 V	5 V	**PASS**
C9	100n	50 V	5 V	**PASS**	C19	22u	25 V	5 V	**PASS**
C10	47n	63 V	5 V	**PASS**	C20	47u	25 V	5 V	**PASS**

Note: the LM386N-1 operating range is 4–12 V, so use a *regulated* 9 V adapter.

8 PCB

136 × 98 mm, two layers, 1.6 mm FR-4, ENIG finish (the keys are exposed copper). Routed with Freerouting; the ground net is carried by copper pours on both layers, with the pour kept out of the keyboard area. Statistics: 81 footprints, 617 track segments, 3 vias, 2/2 pours filled.

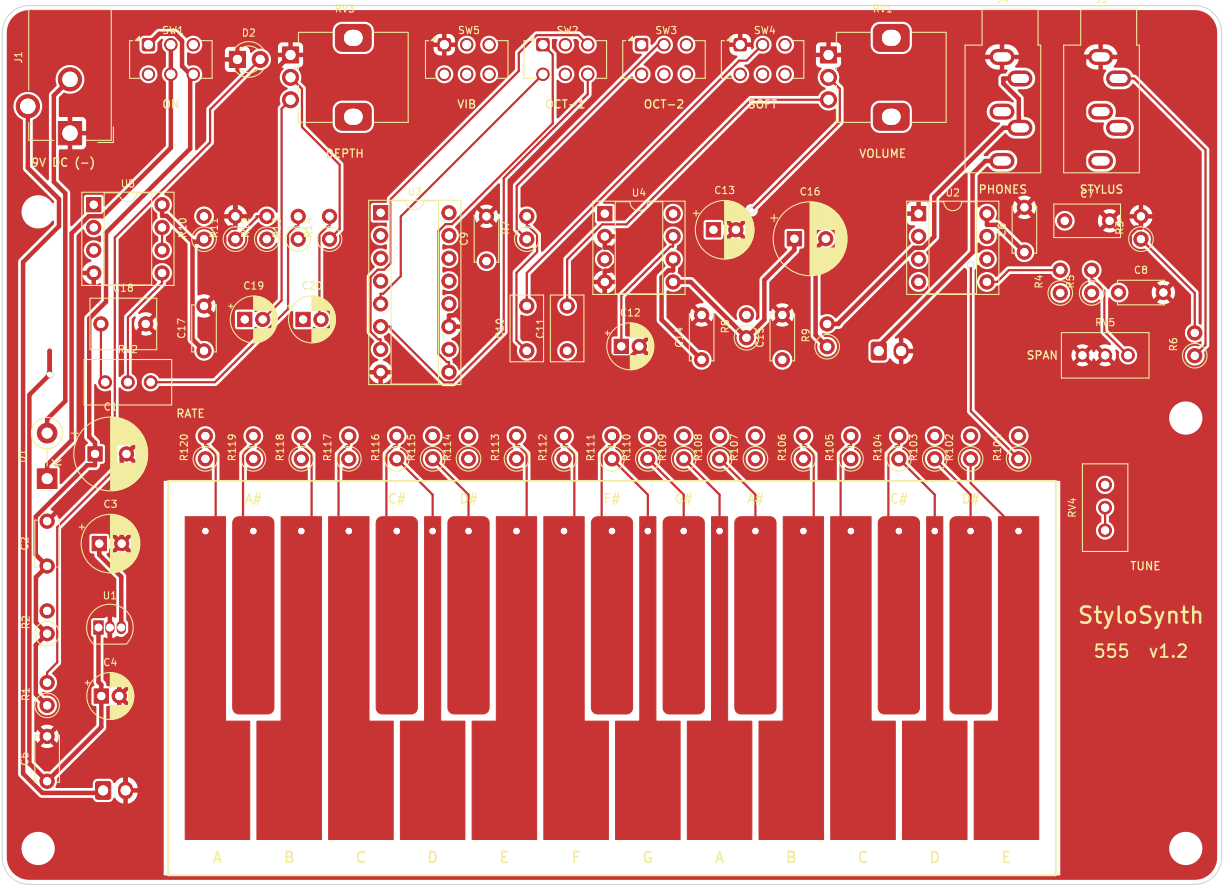
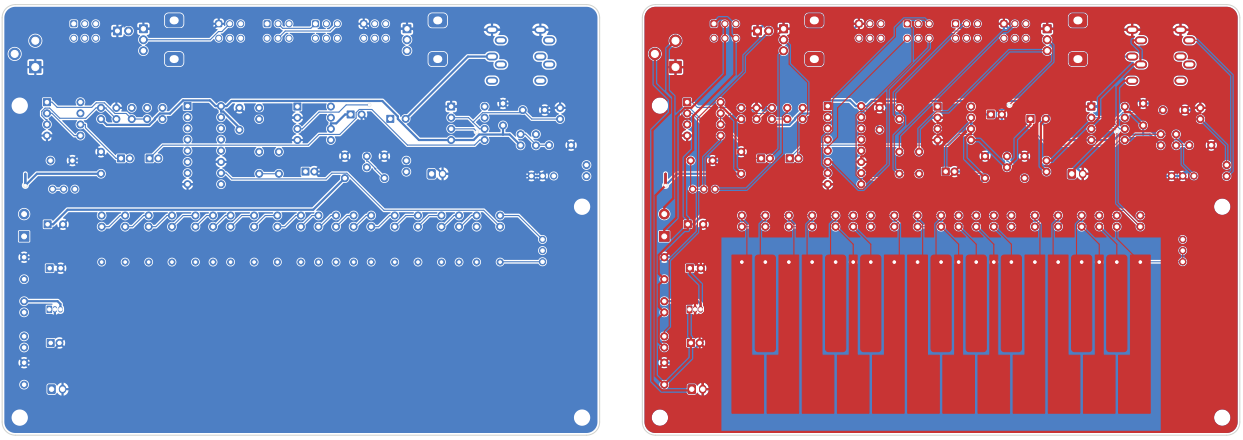


Figure 10: Front copper and silkscreen. The 20 piano-style keys are exposed ENIG pads; the ladder resistors sit in a row directly above the key vias.



(a) Back copper

(b) Both copper layers

Figure 11: PCB layers.

Table 6: PCB verification.

Check	Result	Status
DRC errors (incl. unconnected items)	0	PASS
DRC warnings (cosmetic)	silk_overlap 23, silk_over_copper 5, silk_edge_clearance 2, track_dangling 2	–
Schematic ↔ PCB pad nets	0 mismatches / 210 pins	PASS
Min. track width	Default 0.25 mm, Ground 0.50 mm, Power 0.50 mm	–
Decoupling U2 pin 8 (+5V)	C6 (100n) at 6.0 mm	PASS
Decoupling U3 pin 16 (+5V)	C9 (100n) at 6.8 mm	PASS
Decoupling U4 pin 6 (+9V)	C13 (100u) at 5.6 mm	PASS
Decoupling U5 pin 8 (+5V)	C17 (100n) at 17.0 mm	WARN
Decoupling U1 pin 1 (+5V)	C4 (10u) at 7.6 mm	PASS

Finding: U5 pin 8 (+5V): nearest decoupling cap C17 is 17.0 mm away (target < 12 mm). Move C17 next to the supply pin in the next revision. Severity is low: U5 only runs the vibrato LFO, from the LDO-filtered rail.

9 Enclosures

Both enclosures use the PCB as the top face, held at its four M3 holes. The battery and speaker sit underneath, and all jacks and switches stand above the enclosure edge, so no cut-outs are needed. A geometric fit check confirms the battery, speaker, bosses and pillars clear the back-side parts and THT leads by at least 1 mm. **Result:** fit check: PASS (battery, speaker, bosses, pillars clear of back-side parts and leads).

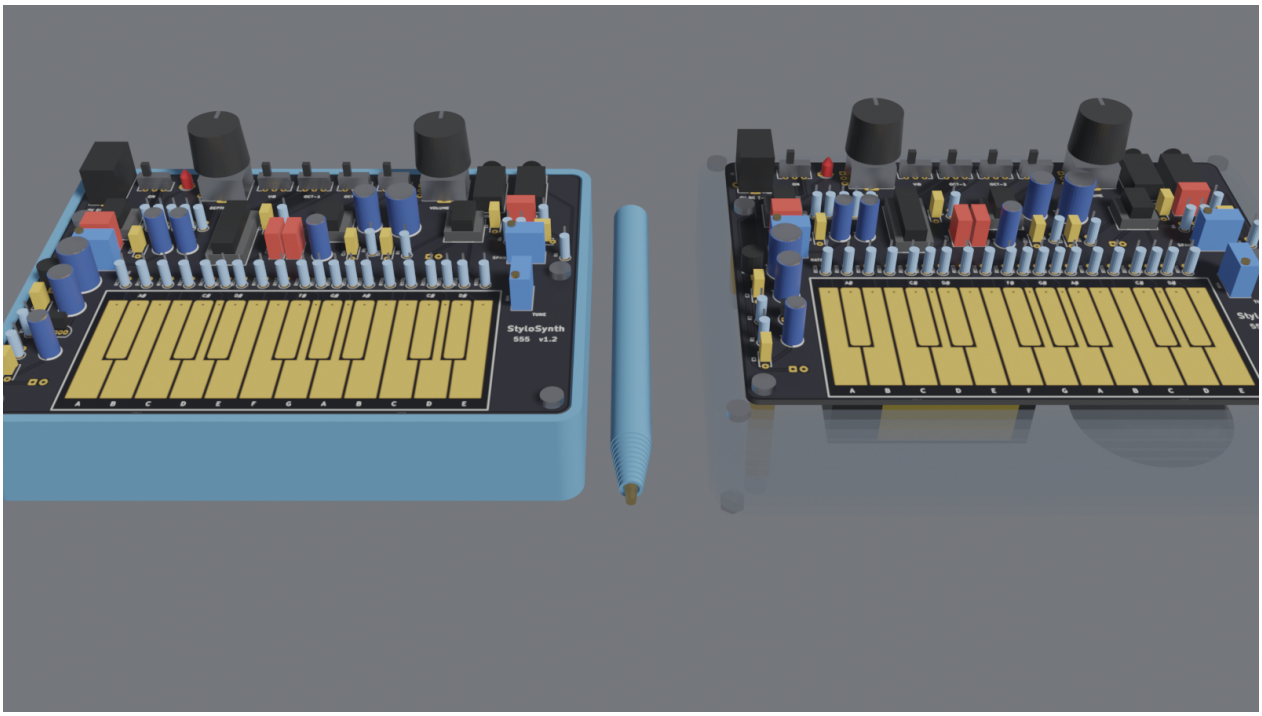
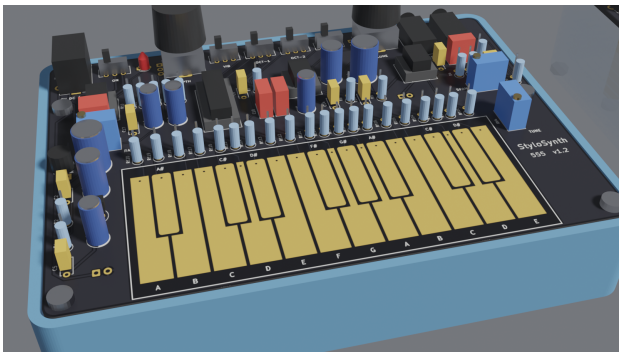
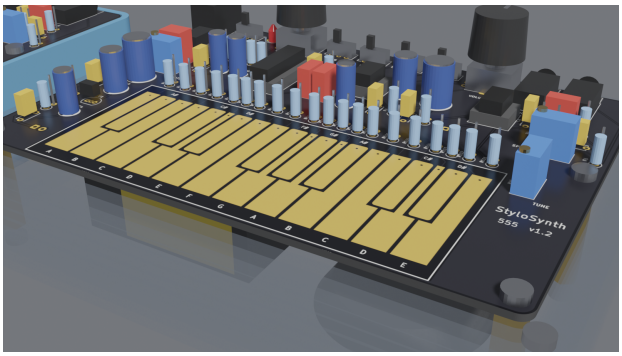


Figure 12: 3D-printed tray with stylus (left) and laser-cut acrylic stack (right), modelled in Blender with the textured PCB and components.



(a) Printed tray: PETG, 141.8×103.8×25.6 mm, ≈62 cm^3



(b) Acrylic: 1 base + 8 rings, 3 mm, 151×113 mm

Figure 13: Enclosure variants.

10 Verification summary

Table 7: Automated verification.

#	Check	Method	Result
1	Circuit behaviour	ngspice, behavioural models	**PASS**
2	Tuning accuracy	2000-build tolerance analysis + calibration	**PASS**
3	Schematic vs source	netlist round-trip	**PASS**
4	Electrical rules	static ERC on symbol pin types	**PASS**
5	Power and ratings	hand calculations	**PASS**
6	PCB	DRC, parity, decoupling	**PASS**
7	Mechanical fit	geometry check	**PASS**

Not yet verified

- Jack and switch pin behaviour (SJ1-3535NG speaker mute, PJ-102AH battery break, JS202011 slider sense), assumed from KiCad symbols. Check against the datasheets and on a breadboard.
- Real IC behaviour with vendor SPICE models (TLC555 output resistance, LM386 stability and hum).
- KiCad GUI ERC (Inspect → Electrical Rules Checker).
- Physical footprint fit: print the PCB at 1:1 and test-fit the pots, switches, jacks and trimmers.
- Stylus-lift clicks, ESD, finger hum, and 5 V rail ripple at full volume, on a prototype.

A Bill of materials

Table 8: BOM (estimated prices, total $66.22). Full detail in `bom/bom.csv`.

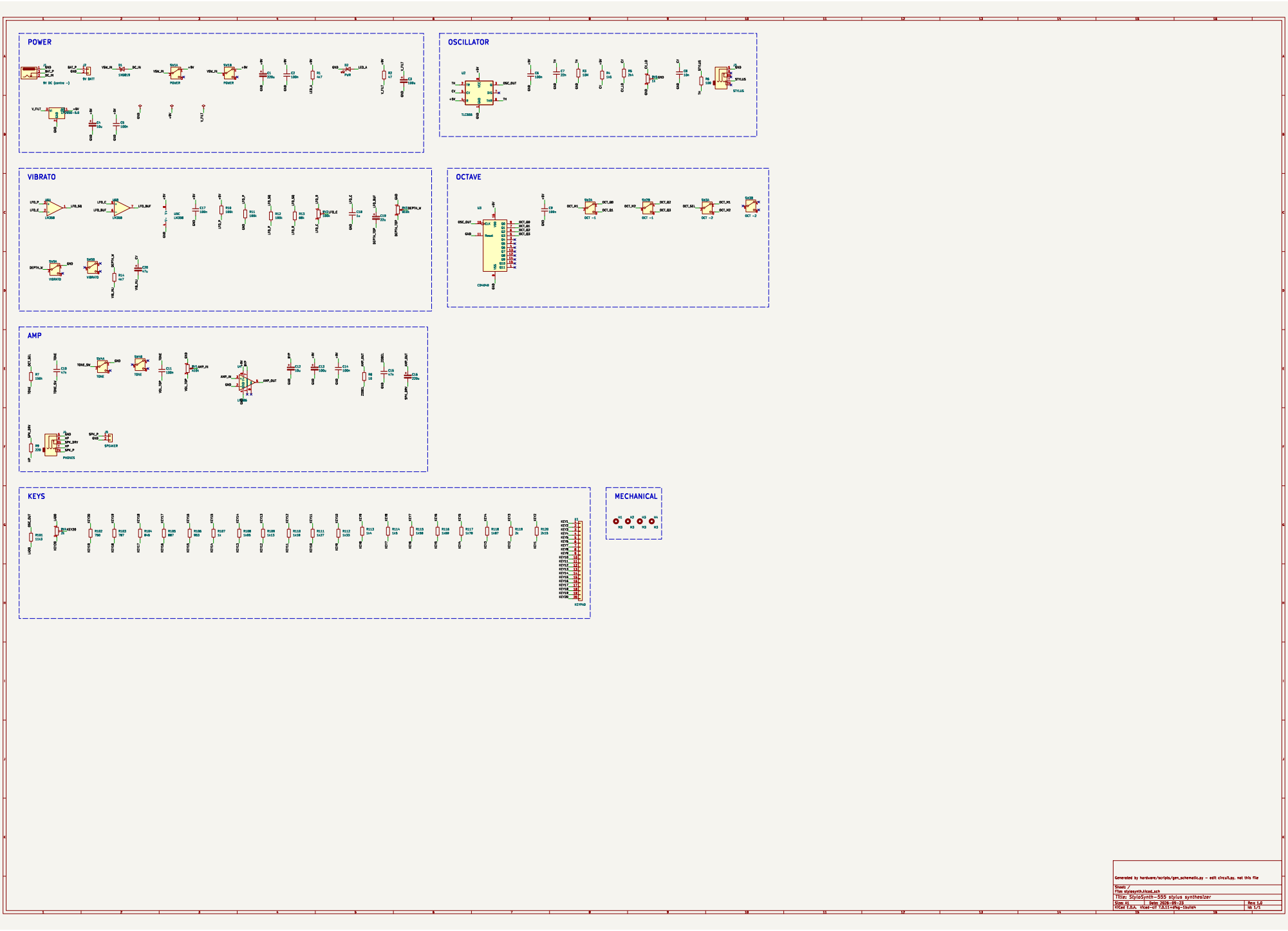
Qty	Refs	Value	Mfr	MPN / description	USD
2	C1, C16	220u	Panasonic	EEU-FC1C221 (220uF 16V, 8 mm, 3.5 mm pitch)	0.90
6	C2, C5, C6, C9, C14, C17	100n	KEMET	C322C104K5R5TA	1.50
2	C3, C13	100u	Panasonic	EEU-FC1C101 (100uF 16V, 6.3 mm, 2.5 mm pitch)	0.70
2	C4, C12	10u	Panasonic	EEU-FC1E100 (10uF 25V, 5 mm, 2 mm pitch)	0.50
1	C7	22n	WIMA	MKP2 22nF 100V 2.5% 5 mm (polypropylene)	0.60
1	C8	10n	KEMET	C322C103K5R5TA	0.25
1	C10	47n	WIMA	MKS2 47nF 63V 5 mm	0.35
1	C11	100n	WIMA	MKS2 100nF 63V 5 mm	0.35
1	C15	47n	KEMET	C322C473K5R5TA	0.25
1	C18	1u	WIMA	MKS2 1uF 63V 5 mm	0.60
1	C19	22u	Panasonic	EEU-FC1E220 (22uF 25V, 5 mm, 2 mm pitch)	0.30
1	C20	47u	Panasonic	47uF 16V, 5 mm dia, 2 mm pitch (FC/FR series)	0.30
1	D1	1N5819	onsemi / Vishay	1N5819	0.20
1	D2	PWR	Kingbright	WP710A10ID (3 mm red diffused)	0.25
1	J1	9V DC (centre -)	Same Sky (CUI)	PJ-102AH (5.5x2.1 mm)	0.90

Qty	Refs	Value	Mfr	MPN / description	USD
1	J2	9V BATT	JST	B2B-XH-A(LF)(SN)	0.15
1	J3	STYLUS	Same Sky (CUI)	SJ1-3535NG	1.30
1	J4	PHONES	Same Sky (CUI)	SJ1-3535NG	1.30
1	J5	SPEAKER	JST	B2B-XH-A(LF)(SN)	0.15
2	R1, R14	4k7	Yageo / generic	4k7 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.06
1	R2	47	Yageo / generic	47 metal film 1% 0.25 W (e.g. Yageo MFR-25 …	0.03
1	R3	10M	Yageo / generic	10M metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
2	R4, R114	1k5	Yageo / generic	1k5 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.06
1	R5	2k4	Yageo / generic	2k4 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R6	100	Yageo / generic	100 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R7	150k	Yageo / generic	150k metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R8	10	Yageo / generic	10 metal film 1% 0.25 W (e.g. Yageo MFR-25 …	0.03
1	R9	220	Yageo / generic	220 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
3	R10, R11, R12	100k	Yageo / generic	100k metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.09
1	R13	68k	Yageo / generic	68k metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R101	11k3	Yageo / generic	11k3 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R102	750	Yageo / generic	750 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R103	787	Yageo / generic	787 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R104	845	Yageo / generic	845 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R105	887	Yageo / generic	887 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R106	953	Yageo / generic	953 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R107	1k	Yageo / generic	1k metal film 1% 0.25 W (e.g. Yageo MFR-25 …	0.03
1	R108	1k05	Yageo / generic	1k05 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R109	1k13	Yageo / generic	1k13 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R110	1k18	Yageo / generic	1k18 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R111	1k27	Yageo / generic	1k27 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R112	1k33	Yageo / generic	1k33 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R113	1k4	Yageo / generic	1k4 metal film 1% 0.25 W (e.g. Yageo MFR-25…	0.03
1	R115	1k58	Yageo / generic	1k58 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R116	1k69	Yageo / generic	1k69 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R117	1k78	Yageo / generic	1k78 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R118	1k87	Yageo / generic	1k87 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	R119	2k	Yageo / generic	2k metal film 1% 0.25 W (e.g. Yageo MFR-25 …	0.03
1	R120	2k15	Yageo / generic	2k15 metal film 1% 0.25 W (e.g. Yageo MFR-2…	0.03
1	RV1	A10k	Alps Alpine / generic	RK09K series 9 mm vertical, 10k log (A10K),…	1.50
1	RV2	100k	Bourns	3296W-1-104LF	1.60
1	RV3	B10k	Alps Alpine / generic	RK09K series 9 mm vertical, 10k lin (B10K),…	1.50
1	RV4	2k	Bourns	3296W-1-202LF	1.60
1	RV5	1k	Bourns	3296W-1-102LF	1.60
1	SW1	POWER	C&K	JS202011CQN	0.60
1	SW2	OCT -1	C&K	JS202011CQN	0.60
1	SW3	OCT -2	C&K	JS202011CQN	0.60
1	SW4	TONE	C&K	JS202011CQN	0.60
1	SW5	VIBRATO	C&K	JS202011CQN	0.60
1	U1	LP2950-5.0	Texas Instruments	LP2950ACZ-5.0/NOPB	1.00
1	U2	TLC555	Texas Instruments	TLC555CP	1.10
1	U3	CD4040	Texas Instruments	CD4040BE	0.70
1	U4	LM386	Texas Instruments	LM386N-1/NOPB	1.20
1	U5	LM358	Texas Instruments	LM358P	0.45
3	-	-	generic	DIP-8 machined socket	0.90
1	-	-	generic	DIP-16 machined socket	0.40

Qty	Refs	Value	Mfr	MPN / description	USD
1	-	-	generic	8R 1W 40 mm	2.50
1	-	-	generic	9V snap, 150 mm leads	0.40
4	-	-	JST	XHP-2 + SXH-001T-P0.6	0.80
1	-	-	generic	3.5 mm mono plug + cable	2.00
1	-	-	generic	brass tip	0.50
2	-	-	generic	6 mm knob	1.60
4	-	-	generic	Ruthex-style M3x5.7	0.60
4	-	-	generic	ISO 7380 M3x6	0.20
4	-	-	generic	M3x25 F-F standoff	1.00
8	-	-	generic	ISO 7380 M3x6	0.40
4	-	-	generic	ISO 7380 M3x30 + DIN 934	0.40
1	-	-	Ponoko/local laser	see enclosure/out/acrylic/*.dxf	15.00
4	-	-	generic	10 mm bumpon	0.40
1	-	-	JLCPCB/PCBWay	fab order	12.00

B Schematic

The complete generated schematic follows on the next page (landscape, vector: zoom in for detail). The source file is `hardware/kicad/stylosynth.kicad_sch`.


POWER
OSCILLATOR
VIBRATO
OCTAVE
AMP
KEYS
MECHANICAL
Generated by hardware/scripts/gen_schematic.py — edit circuit.py, not this file
Sheet: /
File: stylosynth.kicad_sch
Title: StyloSynth—555 stylus synthesizer
Size: A1
Date: 2026-05-23
KiCad 7.0.1 - KiCad-GIT 7.0.11+dfsg-1build4
Rev: 1.0
Id: 1/1